ZECKATHON • PROBLEM STATEMENT 02

Risk & Fraud: An Early-Warning System That Would Have Caught IL&FS

Machine Learning | Corporate Credit Risk | Financial Forensics

THE MANDATE

Build an early-warning system (EWS) that identifies listed companies approaching default or potentially concealing financial manipulation at least **four quarters before the event**, while keeping the alert volume low enough for a real credit desk to act on.

01 | Business Case

A mid-sized bank's corporate credit team monitors **800 borrowers** but can review only about **40 borrowers (5%) in depth each quarter**. Indian corporate failures—including IL&FS, DHFL, Jet Airways, Reliance Capital, Future Retail, and Coffee Day Enterprises—raise the question of whether published financials and other public signals can reveal distress early enough to change credit decisions.

THE CORE QUESTION

Can your system produce a ranked quarterly watch-list, explain why each company is flagged, and recover Indian distress cases without overwhelming the team with false alarms?

02 | What Teams Must Deliver

1 Distress model

Train and validate a bankruptcy/distress model on a public benchmark: the UCI Polish Companies Bankruptcy dataset or the Taiwanese Bankruptcy Prediction dataset. Use a time-aware or genuinely out-of-sample split, and document class imbalance and evaluation methodology.

2 Indian case test

Build a panel of at least 10 Indian listed companies that defaulted or entered insolvency, with documented event dates, plus at least 30 matched non-defaulting peers from the same sectors. Extract the required financial ratios from annual reports for the three years before each event.

3 Manipulation screen

Include Beneish M-Score, Piotroski F-Score, and Altman Z-Score (or a learned equivalent). Add at least two non-financial or "soft" signals, such as auditor resignation, rising promoter pledge, related-party loans, credit-rating actions, or delayed filings.

4 Explainability

For every flagged company, show the top three drivers in plain language—using SHAP values or a comparable method. Write explanations as a credit officer would read them, connecting each driver to a specific concern.

5 Operating policy

Define the alert threshold, quarterly ranking method, review workflow, escalation path, and the cost of false positives versus missed defaults. Explain how the team can investigate the 40 highest-priority names from a universe of 800 borrowers.

03 | Data Sources

SOURCE	WHAT TO COLLECT
Benchmark datasets	UCI Machine Learning Repository: Polish Companies Bankruptcy or Taiwanese Bankruptcy Prediction dataset.
Company disclosures	Annual reports, financial statements, shareholding patterns, exchange announcements, and filings from BSE/NSE.
Credit-rating actions	CRISIL, ICRA, and CARE press releases and rating rationales, including downgrades, watch placements, and default classifications.
Insolvency and event dates	IBBI records and documented public announcements to establish default, insolvency admission, or other event dates.

04 | Evaluation Metrics

METRIC	TARGET / REPORTING REQUIREMENT
Recall of Indian defaulters at a 5% alert rate	Higher is better. Report recall at both four-quarter and eight-quarter lead times.
PR-AUC on benchmark hold-out	Report precision–recall AUC on the untouched hold-out set and state the class imbalance.
Expected loss	A missed default costs 20× a false alarm. Choose thresholds to minimise total expected cost and show the calculation.
Leakage control	Use only information that was publicly available before each company’s event date. No post-event data in features, labels, or preprocessing.

05 | What Strong Submissions Demonstrate

A useful quarterly watch-list—not just a high-scoring model. The strongest submissions combine leakage-safe validation, defensible Indian case evidence, clear explanations, and a review policy that fits the bank’s 5% capacity constraint. Be explicit about missing data, survivorship bias, sector effects, and the limits of historical case reconstruction.

Cross-examination standard: Judges will test whether the system could truly have flagged companies before the event, whether the 5% alert budget is respected, and whether the proposed threshold minimises the stated cost of errors.